Kanyon Phillips

CS 454 Midterm

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21 October, 2026

# CS 454 Midterm Project Evidence Log

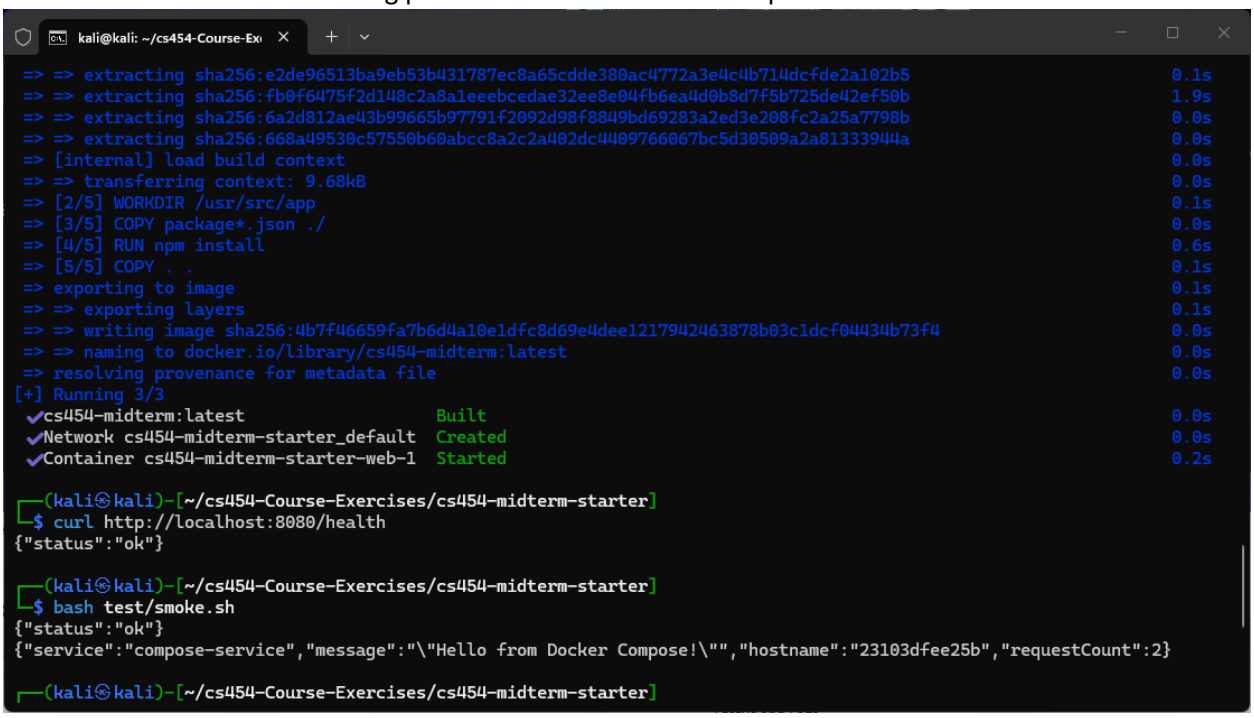
## Part 1:  Small web service

Used starter service on a Kali VM run using Windows Hyper-V manager

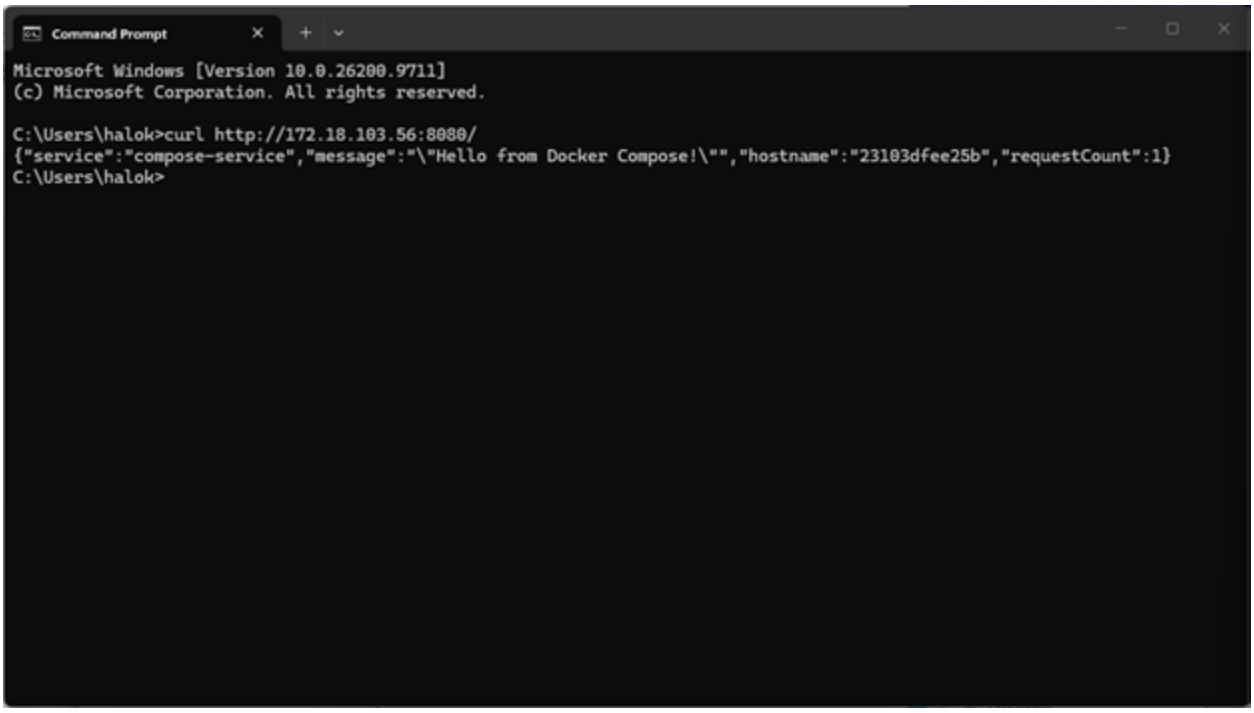
## Part 2: Docker Compose inside a Linux VM

Used Hyper-V Default Switch (NAT), Docker Compose mapped container internal port 8080 to vm port 8080

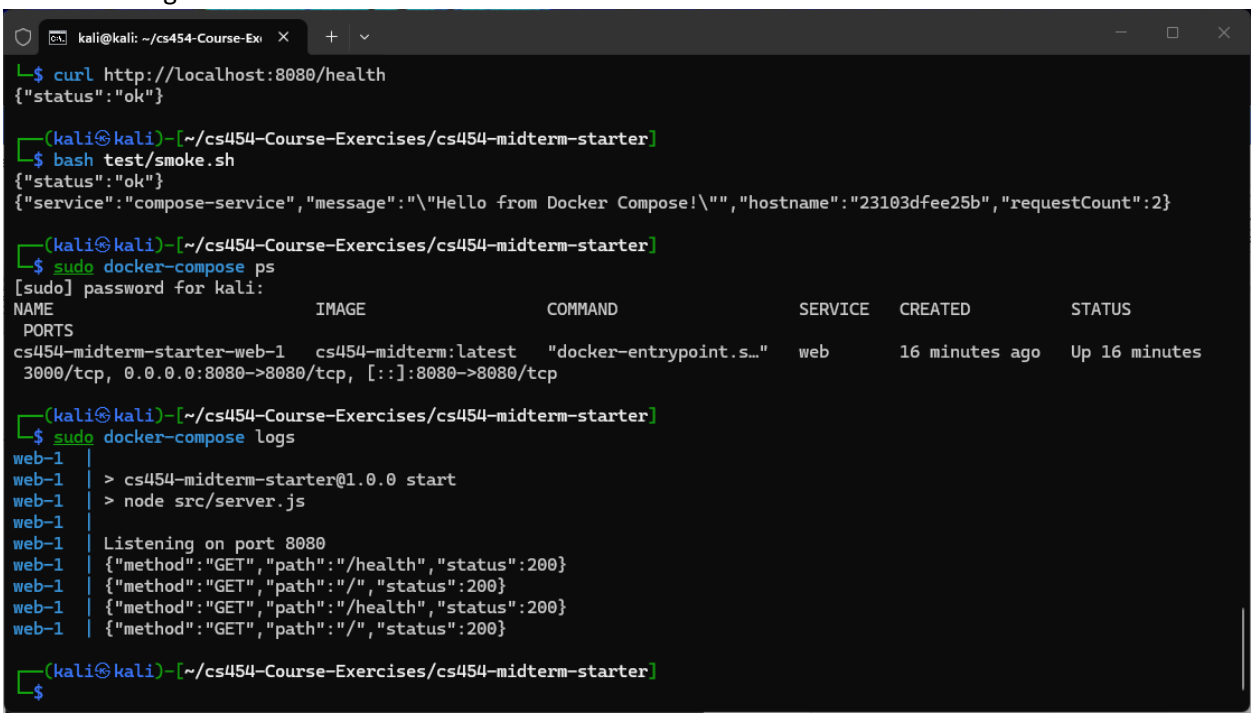
SSH window into kali VM running provided smoke.sh to test endpoints



Local connection to service




Status and Logs

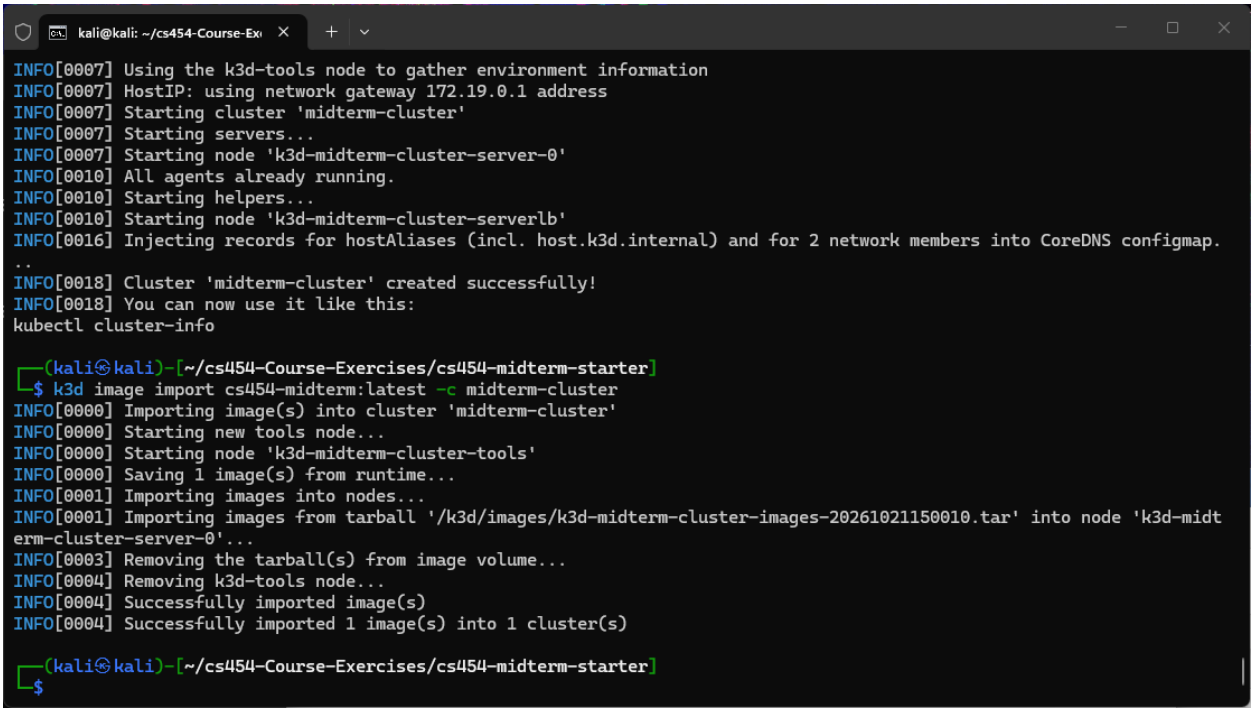



The request count resets when the service is restarted

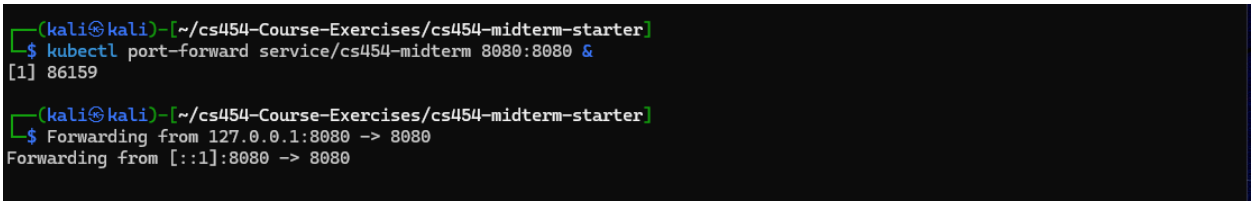
## Part 3: Kubernetes deployment

Kubernetes cluster is running on the Kali VM the Dockerfile was made on

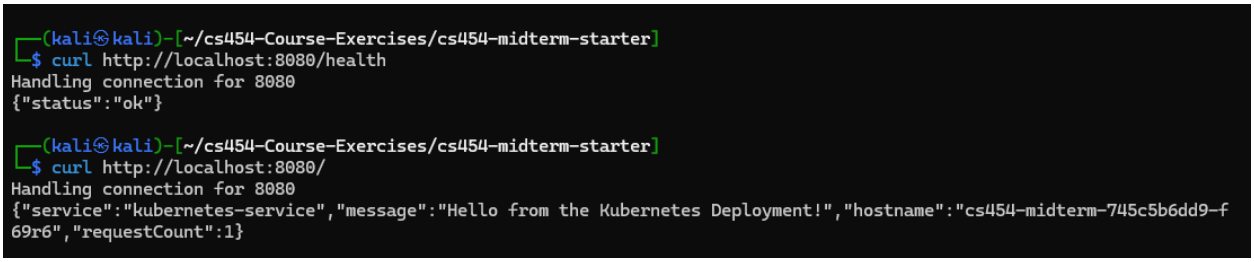
Cluster deployment



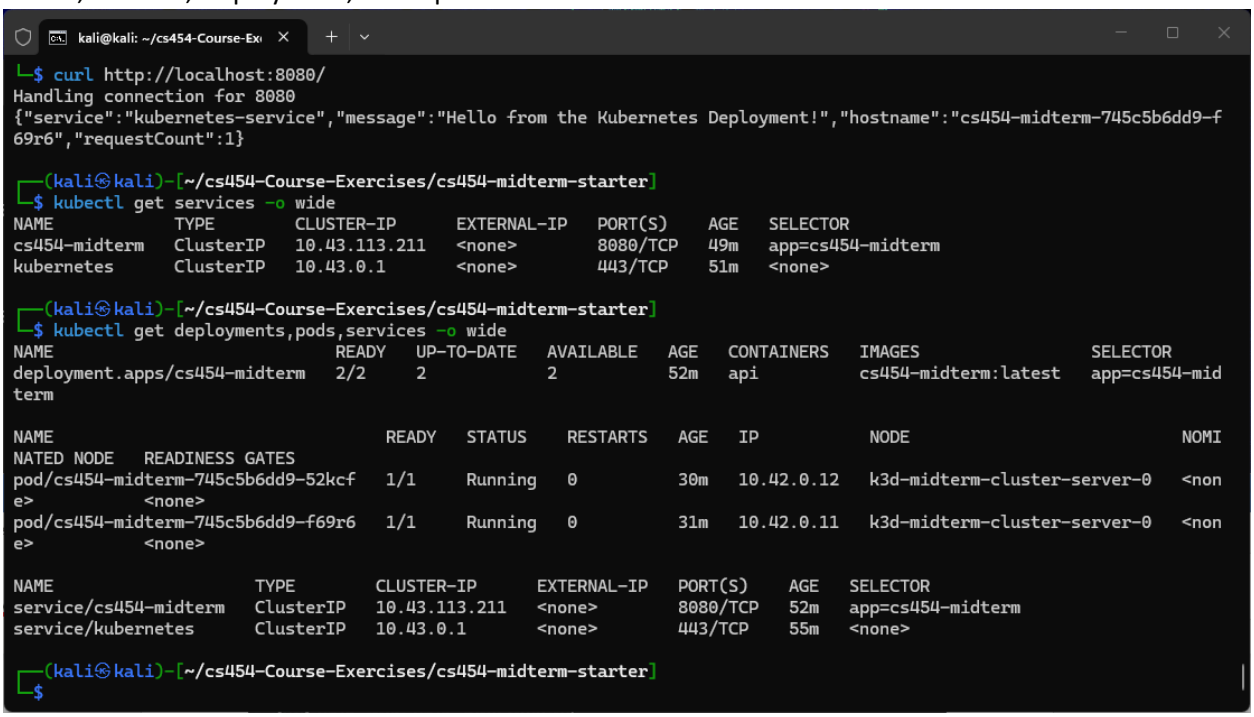

Port forwarding VM port 8080 to Kubernetes



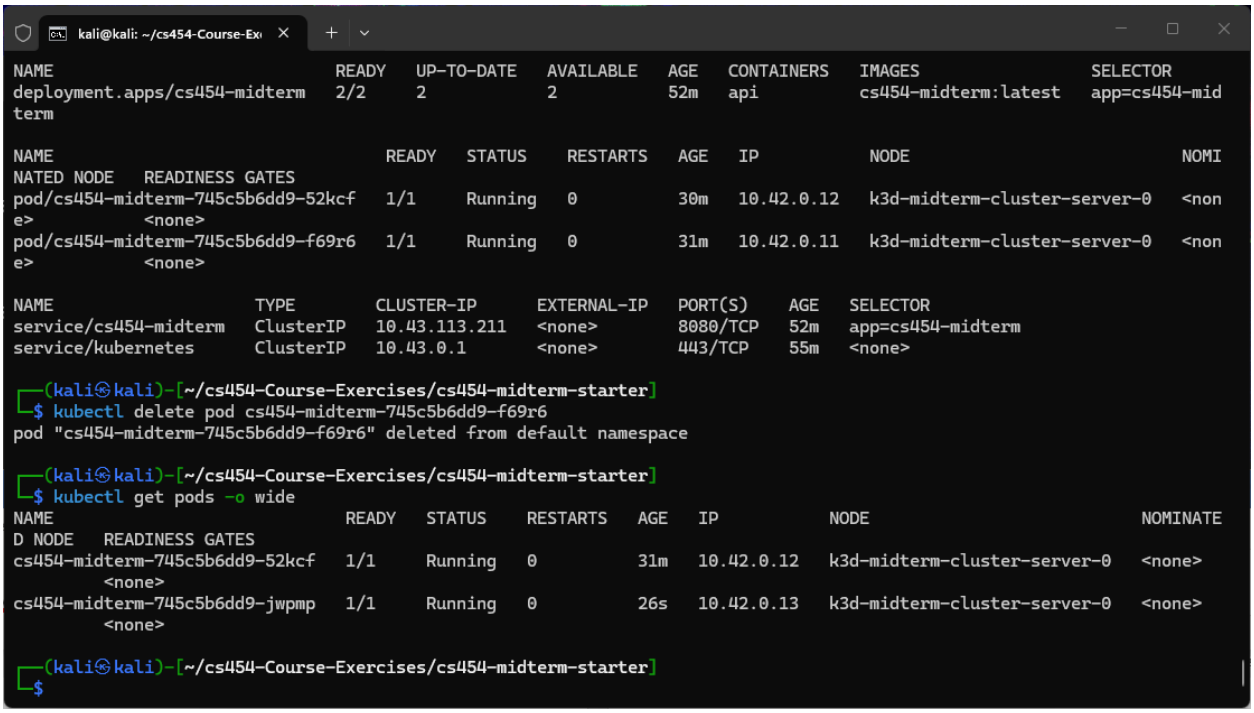

2 requests through the service



Status, services, Deployment, and 2 pods shown

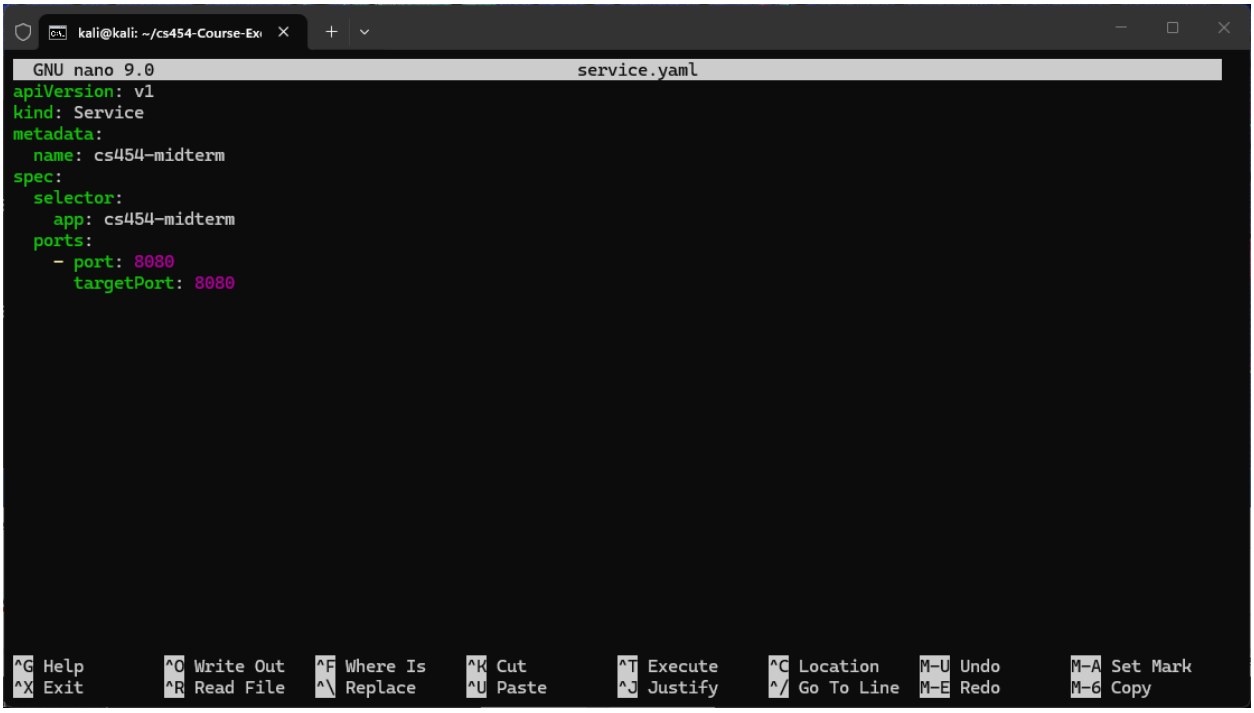



Deleting pod and checking recovery



## Part 4: Controlled Troubleshooting

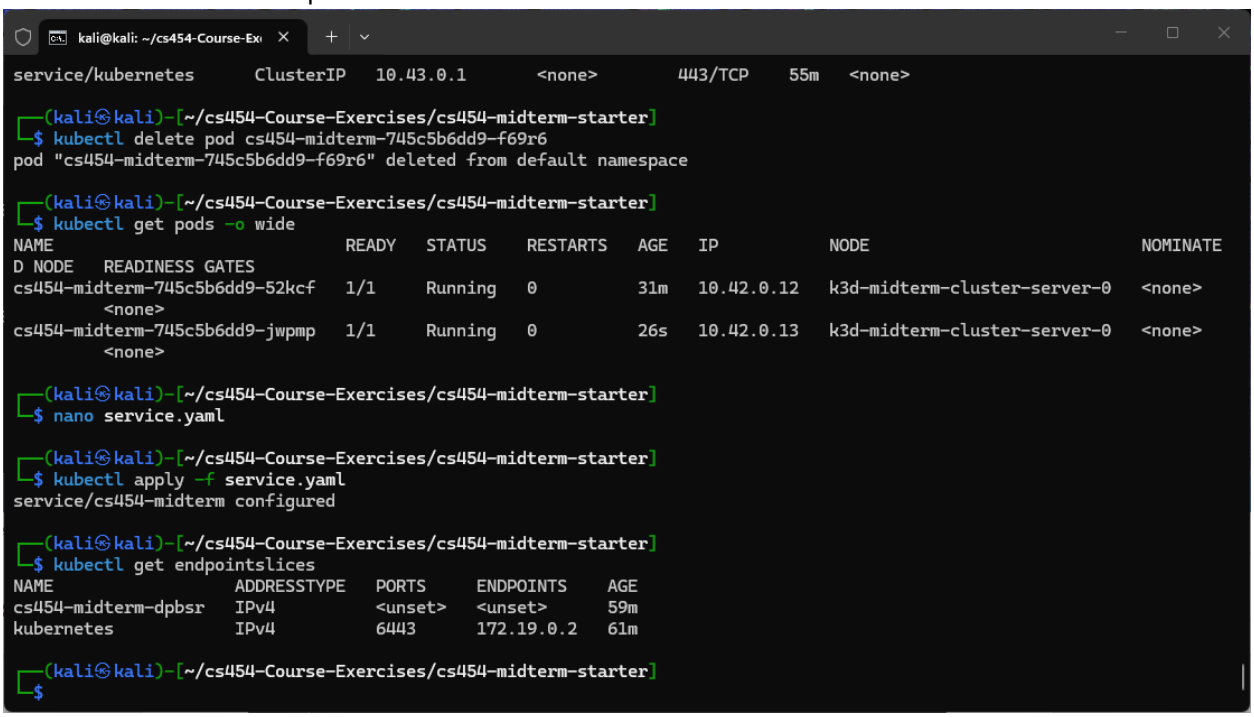
Working service.yaml



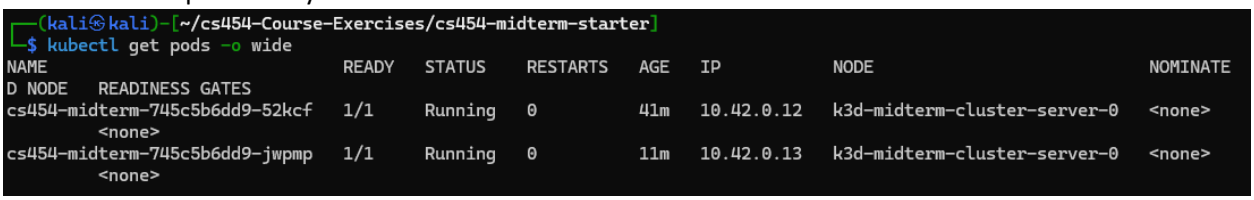

Change



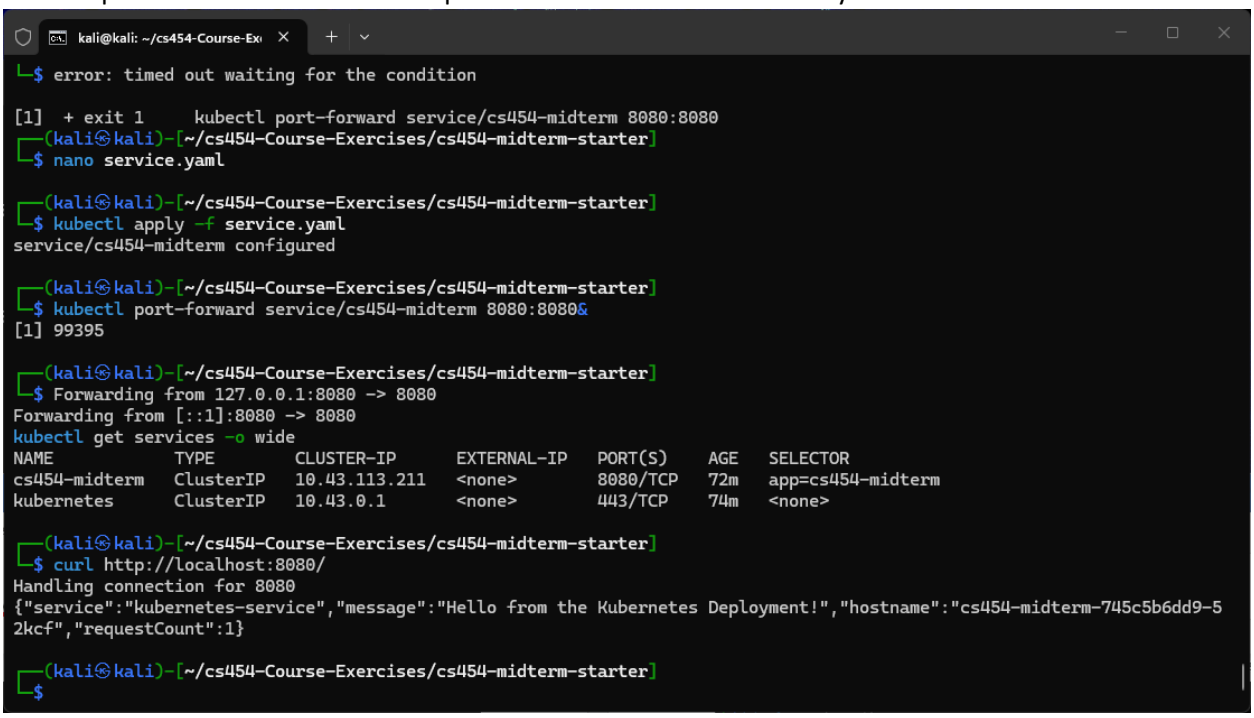
Observation 1: Unset endpoints




Observation 2: pods ready



Explanation, recovery and successful request: The errors are caused by a mismatch between the selector and the pods. The fix is to match the pods with the selector in service.yaml




Explain why healthy application Pods do not, by themselves, prove that a client can reach the application through a Service.: A client's traffic is routed through the service, so if a service has a mismatch with its selector and the pods it will take in the client's traffic, have no idea where to send it, and drop it. During all of that the pods will be perfectly healthy but isolated from the client.